

System Model: Three-Node NTN-ISAC System (Single-Code PMCW)

Transmit and received signal mathematics for a single-waveform (single-code) PMCW joint radar–communication design.

The three-node system consists of:

- **Drone (D):** A mid-air platform with one transmit and one receive antenna. It acts simultaneously as the Joint Radar–Communication (JRC) transmitter, a communication access point for the UE, and a monostatic radar for sensing the moving object.
- **Ground User Equipment (UE):** A single-antenna mobile receiver that receives the downlink communication signal from the drone, and whose position and velocity are estimated by the drone.
- **Moving Object (T):** A passive aerial target (e.g., a drone or UAV) detected and tracked through radar backscatter.

The drone transmits a **single-code PMCW waveform** shared by both the communication and the radar-sensing functions (see [Section 3](#)).

1. Notation and Geometry

Let d_{DU} , d_{DT} , and d_{TU} denote the instantaneous Euclidean distances between the drone and the UE, between the drone and the object, and between the object and the UE, respectively. The corresponding one-way propagation delays are

$$\tau_{DU} = \frac{d_{DU}}{c_0}, \quad \tau_{DT} = \frac{d_{DT}}{c_0}, \quad \tau_{TU} = \frac{d_{TU}}{c_0},$$

where c_0 is the speed of light. The round-trip monostatic sensing delay for the moving object is

$$\tau_{\text{mono}} = \frac{2 d_{DT}}{c_0},$$

and the total propagation delay for the two-hop scattering path $D \rightarrow T \rightarrow U$ is

$$\tau_{DTU} = \tau_{DT} + \tau_{TU} = \frac{d_{DT} + d_{TU}}{c_0}.$$

Let $\mathbf{v}_D$, $\mathbf{v}_U$, and $\mathbf{v}_T$ be the velocity vectors of the drone, UE, and moving object. The radial velocities along each link are projected onto the unit line-of-sight (LOS) vector $\hat{\mathbf{u}}_{ij}$ from node i to node j :

$$v_{r,DU} = (\mathbf{v}_U - \mathbf{v}_D) \cdot \hat{\mathbf{u}}_{DU}, \quad v_{r,DT} = (\mathbf{v}_T - \mathbf{v}_D) \cdot \hat{\mathbf{u}}_{DT}, \quad v_{r,TU} = (\mathbf{v}_U - \mathbf{v}_T) \cdot \hat{\mathbf{u}}_{TU}.$$

With drone carrier wavelength $\lambda_D = c_0/f_{c,D}$, the resulting Doppler shifts are

$$f_{D,DU} = -\frac{v_{r,DU}}{\lambda_D}, \quad f_{D,DT} = -\frac{v_{r,DT}}{\lambda_D}, \quad f_{D,TU} = -\frac{v_{r,TU}}{\lambda_D}.$$

The two-way monostatic Doppler shift for target sensing is $\nu_{\text{mono}} = 2 f_{D,DT}$, and the combined Doppler shift accumulated over the two-hop $D \rightarrow T \rightarrow U$ scattering path is

$$\nu_{DTU} = f_{D,DT} + f_{D,TU}.$$

2. Channel Model

All drone-to-node and node-to-drone links are SISO channels. The channel coefficient between nodes i and j is decomposed as

$$h_{ij} = \sqrt{\beta_{ij}} g_{ij},$$

where β_{ij} is the large-scale power gain (path loss and shadowing) and g_{ij} is the complex small-scale fading coefficient. A distance-dependent path-loss model gives

$$\beta_{ij} = C_{ij} \left(\frac{d_{ij}}{d_0} \right)^{-\alpha_{ij}},$$

with reference channel gain $C_{ij} = G_i G_j (\lambda_D / (4\pi d_0))^2$ at distance d_0 , and α_{ij} the path-loss exponent.

The time-varying channels (including the Doppler phase rotation) are modeled as

$$h_{DU}(t) = \sqrt{\beta_{DU}} g_{DU} e^{j2\pi f_{D,DU} t}, \quad h_{DT}(t) = \sqrt{\beta_{DT}} g_{DT} e^{j2\pi f_{D,DT} t}, \quad h_{TD}(t) = \sqrt{\beta_{TD}} g_{TD} e^{j2\pi f_{D,DT} t}.$$

Under channel reciprocity, $\beta_{TD} = \beta_{DT}$ and $g_{TD} = g_{DT}$.

To account for the scattering path $D \rightarrow T \rightarrow U$ (Section 4), an additional channel between the moving object and the UE is introduced:

$$h_{TU}(t) = \sqrt{\beta_{TU}} g_{TU} e^{j2\pi f_{D,TU} t},$$

where β_{TU} is the large-scale path gain and $g_{TU} \sim \mathcal{CN}(0, 1)$ is the small-scale fading of the $T \rightarrow U$ link.

h_{TU} is an unknown channel. The object's position d_{TU} , velocity $\mathbf{v}_T$, and scattering pattern toward the UE are all unknown a priori (they are precisely what the radar is trying to estimate), so β_{TU} , g_{TU} , and $f_{D,TU}$ cannot be predicted by either the drone or the UE. The $D \rightarrow T \rightarrow U$ interference therefore cannot be cancelled by pre-coding or Doppler pre-compensation at the transmitter.

3. Transmit Signal

The drone transmits a Phase-Modulated Continuous Wave (PMCW) waveform built from a single Maximum-Length Sequence (MLS) of length N_D , shared by both the communication and the radar-sensing functions.

Binary chips and data bits. The raw MLS sequence is generated in binary (logical) form, and the communication payload on pulse m is a raw bit:

$$c[n] \in \{0, 1\}, \quad 0 \leq n < N_D; \quad b_m \in \{0, 1\}, \quad 0 \leq m < M.$$

Polar (BPSK) mapping. Both the chips and the data bit are mapped from the logical alphabet $\{0, 1\}$ to the antipodal **polar** alphabet $\{+1, -1\}$ via the BPSK rule $x \mapsto 1 - 2x$:

$$s_{\text{mls}}[n] = 1 - 2c[n] \in \{+1, -1\}, \quad d_m = 1 - 2b_m \in \{+1, -1\}.$$

In particular $d_m^2 = 1$, a property exploited for radar data stripping (Section 5.1).

Frame structure (preamble + data). Each slow-time block m is divided into two consecutive periods, each of N_D chips and duration $T_m = N_D T_c$, giving a total block duration of $T_{\text{block}} = 2T_m = 2T_{\text{PRI}}$:

Period	Duration	Content	Purpose
Sounding (preamble)	T_m	Pure MLS $s_{\text{mls}}[n]$ (no data modulation, $d_m = +1$)	Channel estimation at UE
Data	T_m	MLS $\times$ polar data symbol d_m	Communication + sensing

The continuous-time baseband transmitted signal for block m is therefore:

$$x_D(t) = \sqrt{P_D} \left[\underbrace{s_{\text{mls}}(t - m T_{\text{block}})}_{\text{sounding period}} + d_m \underbrace{s_{\text{mls}}(t - m T_{\text{block}} - T_m)}_{\text{data period}} \right],$$

where $s_{\text{mls}}(t) = \sum_{n=0}^{N_D-1} s_{\text{mls}}[n] p(t - nT_c)$ and the rectangular chip pulse is

$$p(t) = \begin{cases} 1, & 0 \leq t < T_c, \\ 0, & \text{otherwise.} \end{cases}$$

In discrete time, the two periods are indexed separately:

$$x_{D,\text{sounding}}[n] = \sqrt{P_D} s_{\text{mls}}[n], \quad x_{D,\text{data}}[n, m] = \sqrt{P_D} s_{\text{mls}}[n] d_m, \quad 0 \leq n < N_D.$$

Single-code consequence. Because one code serves both functions, the UE signal and the target echo are **not separable in the code domain**. At the drone receiver they share one matched-filter output and are distinguished only by their **delay and Doppler** — requiring successive interference cancellation (Section 5.1) to keep the strong UE echo from masking the weak target.

4. Received Signal at the UE

Besides the direct drone-to-UE link, the UE also receives a **scattered contribution** over the two-hop path $D \rightarrow T \rightarrow U$: the transmitted signal hits the moving object and is re-radiated toward the UE, acting as uncontrolled interference. The composite received signal is

$$y_U[n, m] = \underbrace{h_{DU} x_D[(n - \ell_{DU})_{N_D}, m] e^{j2\pi f_{D,DU} m T_{\text{PRI}}}}_{\text{direct } D \rightarrow U \text{ link}} + \underbrace{y_{DTU}[n, m]}_{\text{scattered } D \rightarrow T \rightarrow U} + n_U[n, m],$$

where $\ell_{DU} = \text{round}(\tau_{DU} f_s)$ is the integer chip delay, $(\cdot)_{N_D}$ denotes modulo- N_D , $f_s = 1/T_c$, and $n_U[n, m] \sim \mathcal{CN}(0, \sigma_U^2)$ is AWGN.

The scattered term propagates from the drone to the object (delay τ_{DT} , Doppler $f_{D,DT}$, channel h_{DT}), is scattered with complex reflection coefficient η_T , then propagates from the object to the UE (delay τ_{TU} , Doppler $f_{D,TU}$, channel h_{TU}):

$$y_{DTU}[n, m] = \eta_T \sqrt{\beta_{DT} \beta_{TU}} g_{DT} g_{TU} \sqrt{P_D} s_{\text{mls}}[(n - \ell_{DTU})_{N_D}] d_m e^{j2\pi \nu_{DTU} m T_{\text{PRI}}},$$

with $\ell_{DTU} = \text{round}(\tau_{DTU} f_s)$ the total two-hop chip delay and $\nu_{DTU} = f_{D,DT} + f_{D,TU}$ the cumulative Doppler. Substituting the transmit signal, the full received signal is

$$y_U[n, m] = \underbrace{\sqrt{P_D} h_{DU} s_{\text{mls}}[(n - \ell_{DU})_{N_D}] d_m e^{j2\pi f_{D,DU} m T_{\text{PRI}}}}_{\text{desired direct link}} + \underbrace{\eta_T \sqrt{\beta_{DT} \beta_{TU}} g_{DT} g_{TU} \sqrt{P_D} s_{\text{mls}}[(n - \ell_{DTU})_{N_D}] d_m e^{j2\pi \nu_{DTU} m T_{\text{PRI}}}}_{\text{scattered interference (unknown channel)}} + n_U[n, m]$$

The data d_m is recovered by correlating with s_{mls} . The scattered term carries the **same** code delayed by ℓ_{DTU} and Doppler-shifted by ν_{DTU} ; since h_{TU} is unknown it cannot be pre-cancelled and is treated as residual structured noise.

4.1 Data Demodulation at the UE

The UE samples the received signal at chip rate $f_s = 1/T_c$, yielding N_D samples for the sounding period and N_D samples for the data period within each block m .

Step 1 — Delay Estimation and Channel Estimation via Sounding Matched Filter.

During the sounding period of block m , the drone transmits the unmodulated MLS ($d_m = +1$, no data), so the received samples at the UE are

$$y_{U,\text{sound}}[n, m] = h_{DU} \sqrt{P_D} s_{\text{mls}}[n - \ell_{DU}] e^{j2\pi f_{D,DU} m T_{\text{block}}} + n_{U,\text{sound}}[n, m],$$

where $\ell_{DU} = \text{round}(\tau_{DU} f_s)$ is the true chip delay and $T_{\text{block}} = 2 T_{\text{PRI}}$.

Instead of performing frequency-domain deconvolution (least-squares division $\hat{H} = Y/S$), which suffers from severe **noise enhancement** at the DC bin (since $|S_{\text{mls}}[0]|^2 = 1$ is 21 dB weaker than the AC bins where $|S_{\text{mls}}[k_f]|^2 = N_D + 1 = 128$), the UE uses a **Matched Filter (cross-correlation)** in the frequency domain:

$$R_{\text{sound}}[k_f, m] = Y_{U,\text{sound}}[k_f, m] \cdot S_{\text{mls}}^*[k_f],$$

where $S_{\text{mls}}[k_f] = \text{DFT}\{s_{\text{mls}}[n]\}$. Converting back to the time domain:

$$r_{\text{sound}}[n, m] = \text{IDFT}_{k_f}\{R_{\text{sound}}[k_f, m]\}.$$

- **Delay Estimation (Timing Synchronization):** The UE does not know the true delay ℓ_{DU} in advance. It estimates it by locating the peak of the average cross-correlation power:

$$\hat{\ell}_{DU} = \arg \max_n \frac{1}{M} \sum_{m=0}^{M-1} |r_{\text{sound}}[n, m]|^2.$$

- **Channel Estimation:** At the estimated peak $\hat{\ell}_{DU}$, the correlation output represents the coherent sum over the N_D chips:

$$r_{\text{sound}}[\hat{\ell}_{DU}, m] \approx N_D \cdot h_{DU} \sqrt{P_D} e^{j2\pi f_{D,DU} m T_{\text{block}}} + \tilde{w}_{\text{sound}}[m].$$

Dividing by the code length N_D yields the scalar channel estimate for block m (which naturally captures the path loss, fading, and the slow-time Doppler phase):

$$\hat{h}_{DU}[m] = \frac{r_{\text{sound}}[\hat{\ell}_{DU}, m]}{N_D} \approx h_{DU} \sqrt{P_D} e^{j2\pi f_{D,DU} m T_{\text{block}}}.$$

**Why Matched Filtering (Conjugate Multiplication) is Preferred over Least-Squares (Division)

If the UE estimates the channel response using frequency-domain division (Least-Squares deconvolution):

$$\hat{H}_{\text{LS}}[k_f, m] = \frac{Y_{U,\text{sound}}[k_f, m]}{S_{\text{mls}}[k_f]} = H_{DU}[k_f, m] + \frac{W_{\text{sound}}[k_f, m]}{S_{\text{mls}}[k_f]}$$

Taking the IDFT to return to the time domain yields the estimate $\hat{h}_{\text{LS}}[n, m]$. Let's compare the noise variance in the time-domain peak bin for both estimators:

1. Matched Filter Estimator (Conjugate Multiplication):

The matched filter output in the time domain is:

$$r_{\text{sound}}[n, m] = \sum_{j=0}^{N_D-1} y_{U,\text{sound}}[j, m] \cdot s_{\text{mls}}^*[(j-n)_{N_D}]$$

Dividing the peak at the true delay ℓ_{DU} by N_D yields the estimate $\hat{h}_{\text{MF}}[m] = h_{DU} \sqrt{P_D} e^{j2\pi f_{D,DU} m T_{\text{block}}} + \tilde{w}_{\text{MF}}[m]$. Since the time-domain noise samples $n_U[n]$ are i.i.d. with variance σ_U^2 , and $|s_{\text{mls}}[j]|^2 = 1$, the noise variance of the matched-filter estimate is:

$$\text{Var}(\tilde{w}_{\text{MF}}[m]) = \text{Var}\left(\frac{1}{N_D} \sum_{j=0}^{N_D-1} n_U[j] s_{\text{mls}}^*[j - \ell_{DU}]\right) = \frac{1}{N_D^2} \sum_{j=0}^{N_D-1} \sigma_U^2 = \frac{\sigma_U^2}{N_D}$$

2. Least-Squares Estimator (Division):

For the LS deconvolution, the frequency-domain noise term is $W_{\text{sound}}[k_f]/S_{\text{mls}}[k_f]$. By Parseval's theorem, the noise variance in the time domain is the average of the frequency-domain variances:

$$\text{Var}(\tilde{w}_{\text{LS}}[m]) = \frac{1}{N_D^2} \sum_{k_f=0}^{N_D-1} \frac{\text{Var}(W_{\text{sound}}[k_f, m])}{|S_{\text{mls}}[k_f]|^2} = \frac{1}{N_D^2} \left[\frac{N_D \sigma_U^2}{1} + (N_D - 1) \frac{N_D \sigma_U^2}{N_D + 1} \right] \approx \frac{2\sigma_U^2}{N_D}$$

This shows that the Least-Squares deconvolution (division) estimator has **twice the noise variance** (a 3 dB SNR penalty) compared to the Matched Filter estimator, because division amplifies the noise at the weak DC frequency bin where $|S_{\text{mls}}[0]|^2 = 1 \ll 128$. For this reason, matched filtering is mathematically the optimal SNR estimator.

Step 2 — Matched Filtering of the Data Period.

During the data period of block m , the received signal (ignoring the scattered interference for clarity) is

$$y_{U,\text{data}}[n, m] = h_{DU} \sqrt{P_D} s_{\text{mls}}[n - \ell_{DU}] d_m e^{j2\pi f_{D,DU} m T_{\text{block}}} + n_{U,\text{data}}[n, m].$$

The UE correlates the received data signal with the reference code S_{mls} to compress the energy:

$$r_{\text{data}}[n, m] = \text{IDFT}_{k_f} \{Y_{U,\text{data}}[k_f, m] \cdot S_{\text{mls}}^*[k_f]\}.$$

Slicing this matrix at the estimated delay bin $\hat{\ell}_{DU}$ yields the despread peak:

$$z_{\text{mf}}[m] = r_{\text{data}}[\hat{\ell}_{DU}, m] \approx N_D \cdot h_{DU} \sqrt{P_D} d_m e^{j2\pi f_{D,DU} m T_{\text{block}}} + \tilde{w}_{\text{data}}[m].$$

Step 3 — Scalar Channel Equalization.

To cancel the channel gain h_{DU} , the transmit power scaling, and the slow-time Doppler phase rotation simultaneously, the despread data peak is divided by the channel estimate $\hat{h}_{DU}[m]$:

$$z_{\text{eq}}[m] = \frac{z_{\text{mf}}[m]}{\hat{h}_{DU}[m]} \approx N_D \cdot d_m + \tilde{w}_{\text{eq}}[m].$$

Step 4 — BPSK Symbol Decision.

The polar symbol estimate is obtained by slicing the real part of the equalized peak:

$$\hat{d}_m = \text{sign}(\text{Re}\{z_{\text{eq}}[m]\}) \in \{+1, -1\}.$$

The decoded binary bit is then recovered via the inverse BPSK map:

$$\hat{b}_m = \frac{1 - \hat{d}_m}{2} \in \{0, 1\}.$$

The resulting **post-equalization SINR** at the UE is

$$\text{SINR}_U = \frac{P_D \beta_{DU} |g_{DU}|^2}{P_D |\eta_T|^2 \beta_{DT} \beta_{TU} |g_{DT}|^2 |g_{TU}|^2 |\rho(\ell_{DTU})|^2 + \frac{\sigma_U^2}{N_D}},$$

where $\rho(\ell) = \frac{1}{N_D} \sum_n s_{\text{mls}}[(n - \ell)_{N_D}] s_{\text{mls}}^*[n]$ is the MLS autocorrelation ($|\rho(\ell)| = 1/N_D$ for $\ell \neq 0$).

- **Numerator:** desired despread signal power.
- **First denominator term:** residual scattered $D \rightarrow T \rightarrow U$ interference, suppressed by $|\rho|^2 \approx 1/N_D^2$ for $\ell_{DTU} \neq \ell_{DU}$. If $\ell_{DTU} = \ell_{DU}$ (target very close to drone), $\rho(0) = 1$ and the term becomes a co-delay co-Doppler jammer — the worst case, not mitigatable by despreading alone.
- **Second denominator term:** thermal noise, reduced by the processing gain N_D through coherent despreading.

4.2 Analytical Derivation of the Theoretical BER (MGF Inversion Method)

To compute the theoretical BER of BPSK under noisy pilot-based channel estimation, we analyze the decision rule from Step 4. Let the despread data peak and the channel estimate be normalized as:

$$X = \frac{z_{\text{mf}}[m]}{N_D} = S d_m + w_{\text{data}}[m]$$

$$Y = \hat{h}_{DU}[m] = S + w_{\text{sound}}[m]$$

where:

- $S = h_{DU} \sqrt{P_D}$ is the complex channel gain.
- $w_{\text{data}}[m], w_{\text{sound}}[m] \sim \mathcal{CN}(0, \sigma_0^2)$ are independent complex Gaussian noises with variance $\sigma_0^2 = \sigma_U^2/N_D$.

Assuming $d_m = +1$ is transmitted, an error occurs if $\text{Re}\{z_{\text{eq}}[m]\} < 0$, which is equivalent to the real part of the product of two complex Gaussians being negative:

$$P_e = P(\text{Re}\{XY^*\} < 0)$$

We define the decision statistic $D = \text{Re}\{XY^*\} = \frac{1}{2}(XY^* + X^*Y)$, which can be expressed in Hermitian quadratic form as:

$$D = \mathbf{v}^H \mathbf{Q} \mathbf{v}$$

where:

$$\mathbf{v} = \begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{CN}(\boldsymbol{\mu}, \mathbf{R}), \quad \boldsymbol{\mu} = \begin{bmatrix} S \\ S \end{bmatrix}, \quad \mathbf{R} = \sigma_0^2 \mathbf{I}_2, \quad \mathbf{Q} = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The Moment Generating Function (MGF) of D is:

$$\phi_D(s) = E[e^{-sD}] = \frac{\exp(-\boldsymbol{\mu}^H \mathbf{R}^{-1}[(\mathbf{I} + s\mathbf{R}\mathbf{Q})^{-1} - \mathbf{I}]\boldsymbol{\mu})}{\det(\mathbf{I} + s\mathbf{R}\mathbf{Q})}$$

Substituting $\mathbf{R}\mathbf{Q} = \frac{\sigma_0^2}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ yields:

1. $\det(\mathbf{I} + s\mathbf{R}\mathbf{Q}) = 1 - s^2 \frac{\sigma_0^4}{4}$
2. $\boldsymbol{\mu}^H \mathbf{R}^{-1}[(\mathbf{I} + s\mathbf{R}\mathbf{Q})^{-1} - \mathbf{I}]\boldsymbol{\mu} = \frac{|S|^2 s}{1 + s^2 \frac{\sigma_0^2}{2}}$

This simplifies the MGF to:

$$\phi_D(s) = \frac{1}{\left(1 - s \frac{\sigma_0^2}{2}\right) \left(1 + s \frac{\sigma_0^2}{2}\right)} \exp\left(-\frac{|S|^2 s}{1 + s \frac{\sigma_0^2}{2}}\right)$$

The error probability is evaluated using bilateral Laplace inversion:

$$P_e = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} \frac{\phi_D(s)}{s} ds$$

where $0 < c < 2/\sigma_0^2$. Closing the contour in the right half-plane (RHP), the integrand is analytic except for a simple pole at $s = 2/\sigma_0^2$ (since the essential singularity is in the left half-plane at $s = -2/\sigma_0^2$).

Applying the Residue Theorem:

$$P_e = -\text{Res}\left(\frac{\phi_D(s)}{s}, s = \frac{2}{\sigma_0^2}\right) = \frac{1}{2} \exp\left(-\frac{|S|^2}{\sigma_0^2}\right) = \frac{1}{2} \exp(-\gamma_{\text{eff}})$$

where the effective SNR is $\gamma_{\text{eff}} = \frac{P_D \beta_{DU} |g_{DU}|^2}{\sigma_U^2 / N_D}$. This is the standard DBPSK-like performance curve resulting from independent, equal-noise pilot and data matched filtering.

4.3 Practical BER Estimation (Statistical Method of Moments)

In practice, if the receiver does not have access to the exact channel gain S or noise variance σ_0^2 , they can be estimated directly from the received samples $y[n] = Ad[n] + w[n]$ using the Method of Moments. By matching the sample moments to the theoretical moments:

- **Second Moment:** $M_2 = E[y^2] = A^2 + \sigma^2$
- **Fourth Moment:** $M_4 = E[y^4] = A^4 + 6A^2\sigma^2 + 3\sigma^4$

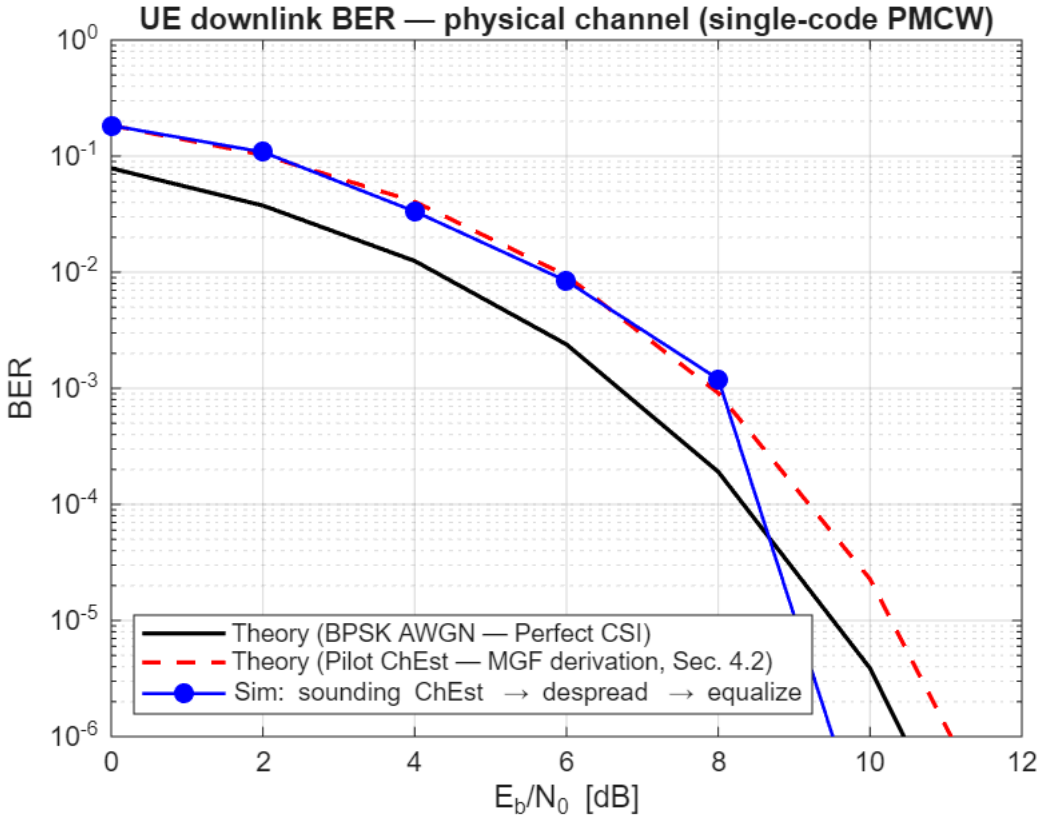
Solving this system yields the estimators:

$$\sigma^2 = M_2 - \sqrt{\frac{3M_2^2 - M_4}{2}}, \quad A^2 = \sqrt{\frac{3M_2^2 - M_4}{2}}$$

The estimated SNR $\hat{\gamma} = A^2/\sigma^2$ is then used to estimate the BER:

$$\hat{P}_e = Q\left(\sqrt{\hat{\gamma}}\right)$$

The resulting BER curves from the simulation are shown below, illustrating the match between the simulated BER and the analytical curve derived via the MGF method:



5. Received Signal at the Drone Receiver (Sensing)

The drone's receive chain collects the backscatter from both the UE (cooperative echo) and the moving object:

$$y_D[n, m] = y_{D,UE}[n, m] + y_{D,T}[n, m] + y_{3hop}[n, m] + n_D[n, m],$$

where $n_D[n, m] \sim \mathcal{CN}(0, \sigma_D^2)$ is receiver noise and $y_{3hop} = y_{DTUD} + y_{DUTD}$ collects the two reciprocal three-hop echoes (defined below).

UE-related echo ($D \rightarrow U \rightarrow D$):

$$y_{D,UE}[n, m] = \eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} s_{\text{mls}}[(n - \ell_{2DU})_{N_D}] d_m e^{j2\pi\nu_{DU} m T_{\text{PRI}}},$$

where η_U is the UE reflection coefficient, $\ell_{2DU} = \text{round}(2\tau_{DU} f_s)$ is the round-trip chip delay, and $\nu_{DU} = 2f_{D,DU}$ the round-trip Doppler.

Moving-object echo ($D \rightarrow T \rightarrow D$):

$$y_{D,T}[n, m] = \eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} s_{\text{mls}}[(n - \ell_{\text{mono}})_{N_D}] d_m e^{j2\pi\nu_{\text{mono}} m T_{\text{PRI}}},$$

where η_T is the complex target reflection coefficient (related to the Radar Cross Section σ_T), $\ell_{\text{mono}} = \text{round}(\tau_{\text{mono}} f_s)$, and $\nu_{\text{mono}} = 2f_{D,DT}$.

Three-hop echoes ($D \rightarrow T \rightarrow U \rightarrow D$ and $D \rightarrow U \rightarrow T \rightarrow D$). Under reciprocity both share the cumulative delay $\ell_{3hop} = \text{round}((\tau_{DT} + \tau_{TU} + \tau_{DU}) f_s)$ and Doppler $\nu_{3hop} = f_{D,DT} + f_{D,TU} + f_{D,DU}$:

$$y_{3hop}[n, m] = 2 \eta_T \eta_U \sqrt{\beta_{DT} \beta_{TU} \beta_{DU}} g_{DT} g_{TU} g_{DU} \sqrt{P_D} s_{\text{mls}}[(n - \ell_{3hop})_{N_D}] d_m e^{j2\pi\nu_{3hop} m T_{\text{PRI}}}.$$

These are **neglected**: they suffer triple free-space path loss ($\propto d^{-6}$) and double scattering loss ($|\eta_T|^2 |\eta_U|^2 \ll 1$), placing them ~ 70 – 80 dB below the noise floor — far weaker than the monostatic target echo ($\propto d^{-4}$, single reflection). Dropping them, the working received signal is

$$y_D[n, m] = \underbrace{\eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} s_{\text{mls}}[(n - \ell_{\text{mono}})_{N_D}] d_m e^{j2\pi\nu_{\text{mono}} m T_{\text{PRI}}}}_{\text{object echo}} + \underbrace{\eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} s_{\text{mls}}[(n - \ell_{2DU})_{N_D}] d_m e^{j2\pi\nu_{DU} m T_{\text{PRI}}}}_{\text{UE echo (interference for sensing)}} + n_D[n, m].$$

Both echoes are spread by the **same** code, distinguished only by delay (ℓ_{mono} vs. ℓ_{2DU}) and Doppler (ν_{mono} vs. ν_{DU}). Since the cooperative UE echo is typically 30–40 dB stronger than the target backscatter, its range sidelobes can mask the weak target — handled by SIC below.

5.1 Sensing Signal Processing at the Drone

A **single-branch** pipeline: one matched filter yields a single joint Range–Doppler map carrying both reflectors; SIC then unmask the weak target before CFAR detection.

Step 1 — BPSK data stripping. The transmitter knows d_m , and both echoes carry it. Since $d_m^2 = 1$, pre-multiplying strips the data modulation, enabling coherent Doppler integration:

$$y'_D[n, m] = y_D[n, m] \cdot d_m.$$

Step 2 — Single matched filtering. Correlate the stripped signal with s_{mls} (efficiently via the FFT):

$$\mathcal{R}[p, m] = \frac{1}{N_D} \sum_{n=0}^{N_D-1} y'_D[n, m] s_{\text{mls}}[(n-p)_{N_D}]^* = \text{IDFT}_{k_f} \{ Y'_D[k_f, m] S_{\text{mls}}^*[k_f] \},$$

with $Y'_D[k_f, m] = \text{DFT}_n \{ y'_D[n, m] \}$ and $S_{\text{mls}}[k_f] = \text{DFT}_n \{ s_{\text{mls}}[n] \}$. The MLS periodic autocorrelation is $R_{cc}[\Delta] = N_D$ at $\Delta = 0$ and -1 otherwise, so this single filter produces **two peaks** plus mutual sidelobe leakage. At the target bin $p = \ell_{\text{mono}}$:

$$\mathcal{R}[\ell_{\text{mono}}, m] \approx \underbrace{\eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} N_D e^{j2\pi\nu_{\text{mono}} m T_{\text{PRI}}}}_{\text{target peak}} + \underbrace{\eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} R_{cc}[\ell_{\text{mono}} - \ell_{2DU}] e^{j2\pi\nu_{DU} m T_{\text{PRI}}}}_{\text{UE sidelobe leakage}} + \tilde{w}_D[\ell_{\text{mono}}, m],$$

and at the UE bin $p = \ell_{2DU}$ (roles swapped):

$$\mathcal{R}[\ell_{2DU}, m] \approx \eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} N_D e^{j2\pi\nu_{DU} m T_{\text{PRI}}} + \eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} R_{cc}[\ell_{2DU} - \ell_{\text{mono}}] e^{j2\pi\nu_{\text{mono}} m T_{\text{PRI}}} + \tilde{w}_D[\ell_{2DU}, m],$$

where $\tilde{w}_D \sim \mathcal{CN}(0, \sigma_D^2/N_D)$ and $|R_{cc}[\Delta]| = 1$ for $\Delta \neq 0$. The MLS sidelobe is only 1 relative to the peak N_D , but because the UE echo is 30–40 dB stronger, its sidelobe at the target bin can still rival or bury the target peak — so matched filtering alone cannot separate the two, motivating SIC (Step 5).

Step 3 — Range estimation. Each peak's delay bin p_{peak} gives the reflector range, with resolution and maximum unambiguous range:

$$R = \frac{c_0 p_{\text{peak}}}{2 f_s}, \quad \Delta R = \frac{c_0 T_c}{2}, \quad R_{\text{max}} = \frac{c_0 N_D T_c}{2} = \frac{c_0 T_{\text{PRI}}}{2}.$$

Step 4 — Doppler FFT (joint Range–Doppler map). A windowed slow-time DFT over the M blocks at each range bin p gives a **single joint map**:

$$\mathcal{X}_{\text{RD}}[p, q] = \sum_{m=0}^{M-1} \mathcal{R}[p, m] w_{\text{slow}}[m] e^{-j\frac{2\pi}{M} q m},$$

with $w_{\text{slow}}[m]$ a window (e.g., Hann) to suppress Doppler sidelobes. The map shows **two peaks**, at $(\ell_{\text{mono}}, q_{\text{mono}})$ for the target and (ℓ_{2DU}, q_{2DU}) for the UE, with peak amplitudes and Doppler

$$A_T = \eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} N_D, \quad A_U = \eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} N_D, \quad \nu_T = 2f_{D,DT}, \quad \nu_U = 2f_{D,DU}.$$

The radial velocity of reflector $i \in \{T, U\}$, the velocity resolution, and the maximum unambiguous velocity are

$$v_{r,i} = \frac{\nu_i \lambda_D}{2}, \quad \nu_i = \frac{q_{\text{peak},i} - M/2}{M T_{\text{PRI}}}, \quad \Delta v = \frac{\lambda_D}{2 M T_{\text{PRI}}}, \quad v_{\text{max}} = \pm \frac{\lambda_D}{4 T_{\text{PRI}}}.$$

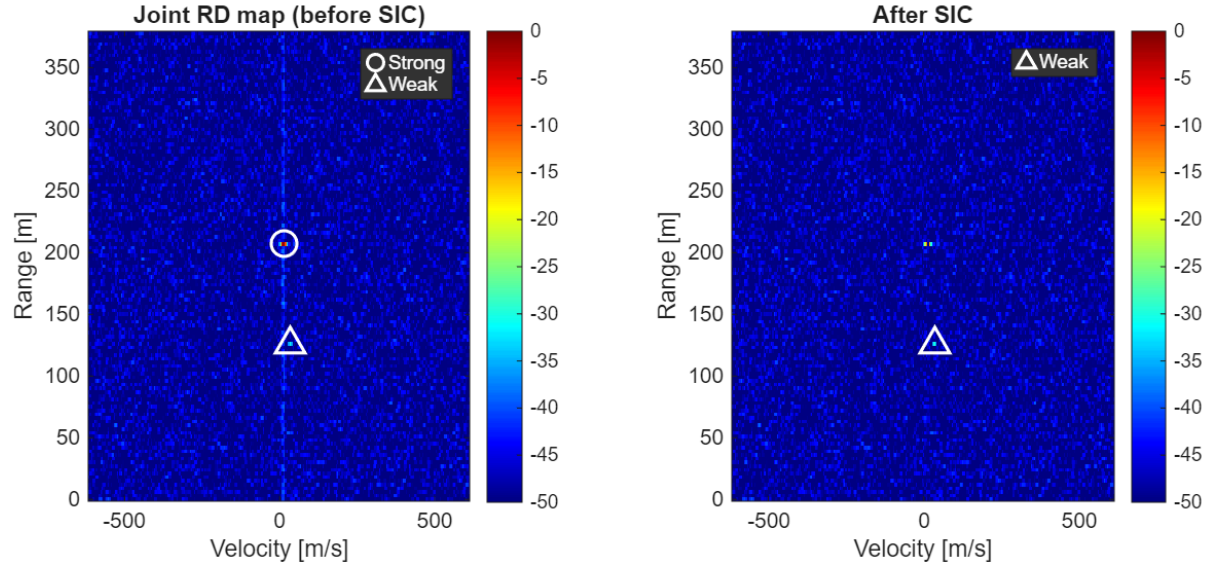
⚠ Warning

Target masking on the joint map. Both reflectors share one heatmap, so the strong UE peak and its sidelobes sit alongside the weak target. If the UE peak — or its sidelobe pedestal — falls in the CFAR training cells of the target's cell-under-test, it inflates the local noise estimate and raises the adaptive threshold, hiding the target (**target masking / signal choking**). This is the fundamental cost of the single-code design, mitigated by SIC.

Step 5 — Successive Interference Cancellation (SIC). Before detecting the weak target:

1. **Detect the dominant UE peak** in $\mathcal{X}_{\text{RD}}[p, q]$ and estimate its delay $\hat{\ell}_{2DU}$, Doppler $\hat{\nu}_{DU}$, and complex amplitude $\hat{A}_U$.
2. **Reconstruct** the data-stripped UE echo $\hat{y}'_{D,\text{UE}}[n, m]$ from $(\hat{\ell}_{2DU}, \hat{\nu}_{DU}, \hat{A}_U)$ and the known code s_{mls} .
3. **Subtract and re-filter:** $y''_D[n, m] = y'_D[n, m] - \hat{y}'_{D,\text{UE}}[n, m]$, then re-correlate with s_{mls} to obtain a cleaned map $\mathcal{X}'_{\text{RD}}[p, q]$ in which the UE peak and sidelobes are largely removed, revealing the target.

The resulting joint Range-Doppler maps before and after SIC are shown below:



Reflector identification. Since both peaks live on one map, the drone cannot label them by code. It cross-references the estimated peak coordinates with the UE's communication telemetry / Doppler feedback (Section 4.1) to identify the UE trajectory, classifying the remaining peak as the passive object.

Step 6 — 2D CA-CFAR detection. On the cleaned map $\mathcal{X}'_{RD}[p, q]$, Cell-Averaging CFAR tests each pixel against an adaptive threshold:

$$\text{Decision}(p, q) = \begin{cases} 1 & |\mathcal{X}'_{RD}[p, q]|^2 > V_{th}(p, q), \\ 0 & \text{otherwise,} \end{cases} \quad V_{th}(p, q) = T \cdot Z(p, q).$$

- $Z(p, q) = \frac{1}{N_{\text{train}}} \sum_{(p', q') \in \mathcal{W}_{\text{train}}} |\mathcal{X}'_{RD}[p', q']|^2$ — local noise power, averaged over the N_{train} training cells in the outer ring (guard cells excluded).
- $T = N_{\text{train}} \left(P_{fa}^{-1/N_{\text{train}}} - 1 \right)$ — threshold multiplier for the target **Probability of False Alarm** P_{fa} (exponential noise model).
- **Guard cells** buffer the CUT so its own power does not leak into $Z(p, q)$ (self-masking).

```

T T T T T T T
T T T T T T T
T T G G G T T
T T G C G T T  ← CUT (C)
T T G G G T T
T T T T T T T
T T T T T T T

```

C = Cell Under Test | G = Guard Cells | T = Training Cells

The **post-processing SNR** at the CUT for reflector $i \in \{T, U\}$ is

$$\text{SNR}_{\text{CUT}, i} = \frac{|A_i|^2 M^2}{\sigma_D^2 N_D M} = \frac{|\eta_i|^2 \beta_{Di}^2 |g_{Di}|^4 P_D N_D M}{\sigma_D^2},$$

where M is the coherent Doppler integration gain and N_D the despreading range processing gain. For the target ($i = T$) this holds after SIC; any residual UE cancellation error adds to the effective noise floor and degrades $\text{SNR}_{\text{CUT}, T}$.